

Appendix — per-structure angles of the adversarial kill-set samples

Aggregated angle = circular mean over the contributing structures (matches the pipeline centroid); spread = circular std (Mardia), degrees. Format per structure: PDBID:chain, res (phi, psi).

KDE-L1 arm

HELIX L-CTC:L-TTG

- 1. L-CTC P30014:158 agg phi=-127.4+/-3.8 psi=-6.7+/-1.2 (n=2) 3V9W:A,159(-124,-6); 4KB1:A,159(-131,-8)
- 2. L-CTC P0AB91:152 agg phi=-135.2+/-0.0 psi=-16.9+/-0.0 (n=1) 1N8F:A,153(-135,-17)
- 3. L-CTC P37355:135 agg phi=-84.2+/-0.8 psi=-14.1+/-2.0 (n=3) 4MXD:A,136(-84,-17); 4MYD:A,136(-84,-14); 4MYS:A,136(-85,-12)
- 4. L-CTC P69506:12 agg phi=-85.6+/-0.0 psi=-13.9+/-0.0 (n=1) 5FNP:A,13(-86,-14)
- 5. L-CTC P0A9Q9:252 agg phi=-81.0+/-0.0 psi=-16.6+/-0.0 (n=1) 1T4B:A,253(-81,-17)
- 6. L-CTC P0A6R0:257 agg phi=-77.6+/-4.0 psi=-3.1+/-8.2 (n=4) 1EBL:A,258(-78,-17); 1HNJ:A,258(-71,+3); 6X7R:A,258(-81,+1); 6X7S:A,258(-81,+1)
- 7. L-CTC P33590:113 agg phi=-77.2+/-3.9 psi=-15.5+/-4.3 (n=14) 1ZLQ:A,92(-75,-14); 2NOO:A,92(-67,-26); 3MVW:A,92(-80,-16); 3MVX:A,92(-82,-13); 3MVZ:A,92(-79,-14); 3MZ9:A,92(-78,-11); 3MZB:A,92(-77,-14); 4I9D:A,92(-77,-18); 5L8D:A,92(-73,-21); 5MWU:A,92(-73,-21); 5ON1:A,92(-80,-13); 5ON5:A,92(-78,-15); 5ON8:A,92(-81,-11); 5ON9:A,92(-80,-10)
- 8. L-CTC P13035:428 agg phi=-81.4+/-0.0 psi=-18.8+/-0.0 (n=1) 2QCU:A,429(-81,-19)

HELIX L-CTC:L-CTG

- 1. L-CTC P0AB91:152 agg phi=-135.2+/-0.0 psi=-16.9+/-0.0 (n=1) 1N8F:A,153(-135,-17)

TURN A-GCG:A-GCT

- 1. A-GCT P27302:54 agg phi=-59.2+/-1.8 psi=-33.4+/-0.8 (n=7) 2R5N:A,55(-58,-34); 2R8O:A,55(-58,-34); 2R8P:A,55(-56,-34); 5HHT:A,55(-60,-32); 6RJC:A,55(-61,-33); 6TJ8:A,55(-60,-34); 6TJ9:A,55(-61,-34)
- 2. A-GCT Q6EZC2:192 agg phi=-57.0+/-0.0 psi=-33.5+/-0.0 (n=1) 2IY9:A,193(-57,-34)
- 3. A-GCT P76578:337 agg phi=-58.3+/-0.0 psi=-31.1+/-0.0 (n=1) 4ZJH:A,338(-58,-31)
- 4. A-GCT D7Y2H5:136 agg phi=-57.9+/-4.9 psi=-37.8+/-6.5 (n=3) 6P7O:A,137(-51,-47); 6P7P:A,137(-61,-33); 6P7Q:A,137(-62,-34)
- 5. A-GCT P08716:565 agg phi=-144.0+/-6.7 psi=-46.3+/-4.0 (n=2) 2FF7:A,566(-151,-50); 2PMK:A,566(-137,-42)
- 6. A-GCT P15005:58 agg phi=-58.9+/-3.8 psi=-36.6+/-3.4 (n=3) 6GCD:A,59(-54,-41); 6GCE:A,59(-60,-36); 6GCF:A,59(-63,-32)
- 7. A-GCT P0A6L2:48 agg phi=-62.5+/-0.7 psi=-34.3+/-0.1 (n=2) 2OJP:A,49(-63,-34); 2PUR:A,49(-62,-34)
- 8. A-GCT P0A6P9:83 agg phi=-57.7+/-0.0 psi=-28.5+/-0.0 (n=1) 2FYM:A,83(-58,-28)

TURN P-CCC:P-CCG

- 1. P-CCC Q68JC9:95 agg phi=-56.2+/-0.0 psi=+131.6+/-0.0 (n=1) 3EOI:A,75(-56,+132)
- 2. P-CCC P78067:186 agg phi=-56.4+/-0.0 psi=+132.8+/-0.0 (n=1) 2WLR:A,167(-56,+133)
- 3. P-CCC P09184:46 agg phi=-55.7+/-0.0 psi=+130.8+/-0.0 (n=1) 1VSR:A,47(-56,+131)
- 4. P-CCC P77489:351 agg phi=-58.5+/-0.0 psi=+134.3+/-0.0 (n=1) 5G5G:C,352(-58,+134)
- 5. P-CCC Q8VQD3:121 agg phi=-55.1+/-0.0 psi=+129.4+/-0.0 (n=1) 1YJ7:A,122(-55,+129)
- 6. P-CCC P00634:40 agg phi=-55.9+/-3.3 psi=+129.0+/-2.7 (n=8) 1ED8:A,19(-58,+123); 1ED9:A,19(-62,+127); 1Y6V:A,19(-56,+130); 1Y7A:A,19(-59,+133); 3BDF:A,19(-54,+131); 3BDG:A,19(-52,+128); 3BDG:B,19(-51,+129); 3TG0:A,19(-55,+130)
- 7. P-CCC P77366:95 agg phi=-58.3+/-0.0 psi=+137.3+/-0.0 (n=1) 4G9B:A,96(-58,+137)
- 8. P-CCC A0A1B3B7F6:173 agg phi=-57.9+/-0.0 psi=+137.2+/-0.0 (n=1) 6VNU:A,174(-58,+137)

torus_p arm

HELIX L-CTC:L-TTG

- 1. L-CTC P77454:302 agg phi=-94.0+/-0.0 psi=-3.5+/-0.0 (n=1) 1U60:A,303(-94,-4)

- 2. L-CTC P29208:237 agg phi=-91.2+/-1.5 psi=-3.7+/-0.5 (n=2) 1FHV:A,238(-93,-3); 1R6W:A,238(-90,-4)
- 3. L-CTC P21179:561 agg phi=-103.9+/-1.8 psi=+0.4+/-1.7 (n=21) 1CF9:A,562(-103,-1); 1P80:A,562(-102,-1); 3P9P:A,562(-105,+0); 3P9Q:A,562(-102,+2); 3PQ2:A,562(-103,-0); 3PQ3:A,562(-102,-1); 3PQ4:A,562(-105,+2); 3PQ5:A,562(-105,+2); 3PQ6:A,562(-108,+3); 3PQ7:A,562(-107,-0); 3PQ8:A,562(-105,+3); 3TTT:A,562(-102,+3); 3TTV:A,562(-104,-2); 3TTW:A,562(-106,-0); 3TTX:A,562(-103,-1); 4ENP:A,562(-100,-1); 4ENR:A,562(-104,+2); 4ENS:A,562(-105,+2); 4ENT:A,562(-104,-2); 4ENU:A,562(-102,-0); 4ENV:A,562(-104,-1)
- 4. L-CTC A0A400L9R1:36 agg phi=-99.2+/-3.8 psi=+5.7+/-3.9 (n=2) 6V2L:A,37(-95,+2); 6V2N:A,37(-103,+10)
- 5. L-CTC P15034:328 agg phi=-87.9+/-0.9 psi=-5.8+/-1.6 (n=6) 2BWS:A,328(-89,-6); 2BWV:A,328(-87,-6); 2BWX:A,328(-89,-3); 2V3X:A,328(-88,-8); 2V3Y:A,328(-86,-5); 2V3Z:A,328(-88,-7)

HELIX L-CTC:L-CTG

- 1. L-CTC P0AB91:152 agg phi=-135.2+/-0.0 psi=-16.9+/-0.0 (n=1) 1N8F:A,153(-135,-17)
- 2. L-CTC A0A400L9R1:36 agg phi=-99.2+/-3.8 psi=+5.7+/-3.9 (n=2) 6V2L:A,37(-95,+2); 6V2N:A,37(-103,+10)
- 3. L-CTC P18956:138 agg phi=-124.2+/-0.8 psi=-32.4+/-1.5 (n=6) 2DBW:A,139(-123,-35); 2DBX:A,139(-123,-31); 2DG5:A,139(-125,-34); 2Z8I:A,139(-124,-31); 2Z8K:A,139(-124,-33); 5B5T:A,139(-125,-32)
- 4. L-CTC P77454:302 agg phi=-94.0+/-0.0 psi=-3.5+/-0.0 (n=1) 1U60:A,303(-94,-4)
- 5. L-CTC P25740:248 agg phi=-89.3+/-1.3 psi=-1.5+/-1.1 (n=2) 2IV7:A,249(-88,-2); 2IW1:A,249(-91,-0)
- 6. L-CTC P29208:237 agg phi=-91.2+/-1.5 psi=-3.7+/-0.5 (n=2) 1FHV:A,238(-93,-3); 1R6W:A,238(-90,-4)
- 7. L-CTC P21179:561 agg phi=-103.9+/-1.8 psi=+0.4+/-1.7 (n=21) 1CF9:A,562(-103,-1); 1P80:A,562(-102,-1); 3P9P:A,562(-105,+0); 3P9Q:A,562(-102,+2); 3PQ2:A,562(-103,-0); 3PQ3:A,562(-102,-1); 3PQ4:A,562(-105,+2); 3PQ5:A,562(-105,+2); 3PQ6:A,562(-108,+3); 3PQ7:A,562(-107,-0); 3PQ8:A,562(-105,+3); 3TTT:A,562(-102,+3); 3TTV:A,562(-104,-2); 3TTW:A,562(-106,-0); 3TTX:A,562(-103,-1); 4ENP:A,562(-100,-1); 4ENR:A,562(-104,+2); 4ENS:A,562(-105,+2); 4ENT:A,562(-104,-2); 4ENU:A,562(-102,-0); 4ENV:A,562(-104,-1)
- 8. L-CTC P0A7J0:77 agg phi=-87.1+/-0.5 psi=-1.2+/-0.3 (n=2) 1G57:A,78(-87,-2); 1G58:A,78(-88,-1)
- 9. L-CTC P31133:163 agg phi=-105.1+/-2.3 psi=+0.3+/-1.5 (n=5) 4JDF:A,164(-109,+3); 6YE0:A,164(-103,+0); 6YE6:A,164(-106,-2); 6YE7:A,164(-105,-0); 6YE8:A,164(-103,+0)
- 10. L-CTC P15034:328 agg phi=-87.9+/-0.9 psi=-5.8+/-1.6 (n=6) 2BWS:A,328(-89,-6); 2BWV:A,328(-87,-6); 2BWX:A,328(-89,-3); 2V3X:A,328(-88,-8); 2V3Y:A,328(-86,-5); 2V3Z:A,328(-88,-7)
- 11. L-CTC P0A6L4:170 agg phi=-82.4+/-1.7 psi=-7.0+/-1.9 (n=4) 2WKJ:A,171(-84,-6); 3LBM:A,171(-82,-9); 3LCG:A,171(-83,-4); 4UUI:A,171(-80,-9)
- 12. L-CTC Q9S5G5:161 agg phi=-84.3+/-1.6 psi=-1.7+/-1.8 (n=4) 2FPR:A,163(-86,+1); 2FPU:A,163(-83,-2); 2FPW:A,163(-83,-4); 2FPX:A,163(-86,-2)
- 13. L-CTC A0A0F3U9S7:506 agg phi=-79.3+/-0.5 psi=-10.2+/-0.8 (n=4) 6SPN:A,506(-79,-10); 6SPO:A,506(-79,-9); 6SPP:A,506(-79,-11); 6SPQ:A,506(-80,-10)
- 14. L-CTC P41409:22 agg phi=-78.3+/-0.0 psi=-11.1+/-0.0 (n=1) 1YOE:A,23(-78,-11)
- 15. L-CTC P24182:443 agg phi=-78.2+/-0.0 psi=-11.9+/-0.0 (n=1) 2W70:A,444(-78,-12)
- 16. L-CTC P69441:212 agg phi=-80.0+/-0.0 psi=-12.4+/-0.0 (n=1) 4X8L:A,213(-80,-12)
- 17. L-CTC A0MMD0:112 agg phi=-76.8+/-0.0 psi=-10.2+/-0.0 (n=1) 1QTX:A,112(-77,-10)
- 18. L-CTC P0AAI9:306 agg phi=-73.2+/-0.0 psi=-16.9+/-0.0 (n=1) 2G2O:A,307(-73,-17)
- 19. L-CTC P0A881:74 agg phi=-111.6+/-8.1 psi=-4.1+/-6.3 (n=10) 1JHG:A,75(-100,-15); 3SSW:N,75(-112,-4); 6EKP:A,75(-117,-1); 6ELB:A,75(-122,+5); 6ELG:A,75(-124,+4); 6ENI:A,75(-113,-2); 6ENN:A,75(-115,-2); 6F7G:A,75(-107,-9); 6F9K:A,75(-99,-14); 6FAL:A,75(-106,-2)
- 20. L-CTC P33590:113 agg phi=-77.2+/-3.9 psi=-15.5+/-4.3 (n=14) 1ZLQ:A,92(-75,-14); 2NOO:A,92(-67,-26); 3MVW:A,92(-80,-16); 3MVX:A,92(-82,-13); 3MVZ:A,92(-79,-14); 3MZ9:A,92(-78,-11); 3MZF:A,92(-77,-14); 4I9D:A,92(-77,-18); 5L8D:A,92(-73,-21); 5MWU:A,92(-73,-21); 5ON1:A,92(-80,-13); 5ON5:A,92(-78,-15); 5ON8:A,92(-81,-11); 5ON9:A,92(-80,-10)

HELIX L-CTC:L-CTT

- 1. L-CTC Q9S5G5:161 agg phi=-84.3+/-1.6 psi=-1.7+/-1.8 (n=4) 2FPR:A,163(-86,+1); 2FPU:A,163(-83,-2); 2FPW:A,163(-83,-4); 2FPX:A,163(-86,-2)
- 2. L-CTC A0MMD0:112 agg phi=-76.8+/-0.0 psi=-10.2+/-0.0 (n=1) 1QTX:A,112(-77,-10)
- 3. L-CTC P0A7J0:77 agg phi=-87.1+/-0.5 psi=-1.2+/-0.3 (n=2) 1G57:A,78(-87,-2); 1G58:A,78(-88,-1)
- 4. L-CTC P0A6L4:170 agg phi=-82.4+/-1.7 psi=-7.0+/-1.9 (n=4) 2WKJ:A,171(-84,-6); 3LBM:A,171(-82,-9); 3LCG:A,171(-83,-4); 4UUI:A,171(-80,-9)
- 5. L-CTC P25740:248 agg phi=-89.3+/-1.3 psi=-1.5+/-1.1 (n=2) 2IV7:A,249(-88,-2); 2IW1:A,249(-91,-0)
- 6. L-CTC A0A0F3U9S7:506 agg phi=-79.3+/-0.5 psi=-10.2+/-0.8 (n=4) 6SPN:A,506(-79,-10); 6SPO:A,506(-79,-9); 6SPP:A,506(-79,-11); 6SPQ:A,506(-80,-10)

- 7. L-CTC P41409:22 agg phi=-78.3+/-0.0 psi=-11.1+/-0.0 (n=1) 1YOE:A,23(-78,-11)
- 8. L-CTC P24182:443 agg phi=-78.2+/-0.0 psi=-11.9+/-0.0 (n=1) 2W70:A,444(-78,-12)
- 9. L-CTC A0A400L9R1:36 agg phi=-99.2+/-3.8 psi=+5.7+/-3.9 (n=2) 6V2L:A,37(-95,+2); 6V2N:A,37(-103,+10)
- 10. L-CTC P69441:212 agg phi=-80.0+/-0.0 psi=-12.4+/-0.0 (n=1) 4X8L:A,213(-80,-12)
- 11. L-CTC P29208:237 agg phi=-91.2+/-1.5 psi=-3.7+/-0.5 (n=2) 1FHV:A,238(-93,-3); 1R6W:A,238(-90,-4)
- 12. L-CTC P15034:328 agg phi=-87.9+/-0.9 psi=-5.8+/-1.6 (n=6) 2BWS:A,328(-89,-6); 2BWV:A,328(-87,-6); 2BWY:A,328(-89,-3); 2V3X:A,328(-88,-8); 2V3Y:A,328(-86,-5); 2V3Z:A,328(-88,-7)
- 13. L-CTC P0AB91:152 agg phi=-135.2+/-0.0 psi=-16.9+/-0.0 (n=1) 1N8F:A,153(-135,-17)
- 14. L-CTC Q9EVJ6:109 agg phi=-65.9+/-0.0 psi=-18.9+/-0.0 (n=1) 3FRQ:A,110(-66,-19)
- 15. L-CTC P0AAI9:306 agg phi=-73.2+/-0.0 psi=-16.9+/-0.0 (n=1) 2G2O:A,307(-73,-17)
- 16. L-CTC P77454:302 agg phi=-94.0+/-0.0 psi=-3.5+/-0.0 (n=1) 1U60:A,303(-94,-4)
- 17. L-CTC P33590:113 agg phi=-77.2+/-3.9 psi=-15.5+/-4.3 (n=14) 1ZLQ:A,92(-75,-14); 2NOO:A,92(-67,-26); 3MVW:A,92(-80,-16); 3MVX:A,92(-82,-13); 3MVZ:A,92(-79,-14); 3MZ9:A,92(-78,-11); 3MZB:A,92(-77,-14); 4I9D:A,92(-77,-18); 5L8D:A,92(-73,-21); 5MWU:A,92(-73,-21); 5ON1:A,92(-80,-13); 5ON5:A,92(-78,-15); 5ON8:A,92(-81,-11); 5ON9:A,92(-80,-10)
- 18. L-CTC P0A7I7:138 agg phi=-69.2+/-0.0 psi=-19.3+/-0.0 (n=1) 2BZ1:A,139(-69,-19)
- 19. L-CTC P75691:250 agg phi=-63.0+/-0.0 psi=-21.7+/-0.0 (n=1) 1UUF:A,251(-63,-22)
- 20. L-CTC P31808:139 agg phi=-67.3+/-0.0 psi=-20.8+/-0.0 (n=1) 3F1L:A,140(-67,-21)

HELIX R-AGG:R-CGA

- 1. R-AGG Q93ET9:278 agg phi=-51.8+/-0.0 psi=-33.8+/-0.0 (n=1) 4DCA:A,279(-52,-34)
- 2. R-CGA P0A7D1:186 agg phi=-70.3+/-0.0 psi=-48.5+/-0.0 (n=1) 2PTH:A,186(-70,-48)
- 3. R-AGG Q8X834:258 agg phi=-55.2+/-0.0 psi=-32.2+/-0.0 (n=1) 4LGJ:A,250(-55,-32)
- 4. R-AGG Q7DI68:193 agg phi=-59.3+/-0.0 psi=-27.4+/-0.0 (n=1) 4M1U:A,172(-59,-27)
- 5. R-CGA Q93ET8:123 agg phi=-60.3+/-0.0 psi=-54.4+/-0.0 (n=1) 6BFF:A,124(-60,-54)
- 6. R-CGA P0A6E4:31 agg phi=-69.5+/-0.0 psi=-46.2+/-0.0 (n=1) 1K92:A,31(-70,-46)
- 7. R-CGA P06721:301 agg phi=-79.0+/-0.4 psi=-39.3+/-2.6 (n=4) 2FQ6:A,302(-79,-44); 2GQN:A,302(-78,-38); 4ITG:A,302(-79,-36); 4ITX:A,302(-79,-39)
- 8. R-AGG P77366:70 agg phi=-62.5+/-0.0 psi=-37.2+/-0.0 (n=1) 4G9B:A,71(-62,-37)
- 9. R-CGA A0A0H2Z2W8:278 agg phi=-74.6+/-0.0 psi=-40.6+/-0.0 (n=1) 6MGC:A,279(-75,-41)
- 10. R-AGG P33362:254 agg phi=-58.9+/-0.0 psi=-31.1+/-0.0 (n=1) 4WEP:A,255(-59,-31)
- 11. R-AGG P03007:134 agg phi=-63.0+/-0.0 psi=-36.5+/-0.0 (n=1) 2GUI:A,135(-63,-36)
- 12. R-CGA P00557:291 agg phi=-67.5+/-2.0 psi=-45.8+/-2.2 (n=3) 3W0O:A,292(-69,-45); 3W0Q:A,292(-65,-49); 3W0S:A,292(-68,-44)

TURN A-GCG:A-GCT

- 1. A-GCG P33590:421 agg phi=-65.9+/-26.2 psi=+149.1+/-16.8 (n=14) 1ZLQ:A,400(-72,+144); 2NOO:A,400(+44,-148); 3MVW:A,400(-71,+148); 3MVX:A,400(-68,+148); 3MVZ:A,400(-65,+143); 3MZ9:A,400(-73,+145); 3MZB:A,400(-73,+146); 4I9D:A,400(-68,+147); 5L8D:A,400(-74,+143); 5MWU:A,400(-74,+143); 5ON1:A,400(-70,+145); 5ON5:A,400(-68,+144); 5ON8:A,400(-68,+145); 5ON9:A,400(-66,+145)
- 2. A-GCG P77366:191 agg phi=-64.0+/-0.0 psi=+148.1+/-0.0 (n=1) 4G9B:A,192(-64,+148)
- 3. A-GCG P08191:254 agg phi=-61.1+/-0.0 psi=+146.5+/-0.0 (n=1) 4XO9:A,234(-61,+146)
- 4. A-GCG Q8X6R9:67 agg phi=-68.2+/-3.1 psi=+152.4+/-3.3 (n=3) 3ZDA:A,68(-65,+152); 3ZDB:A,68(-72,+156); 3ZDC:A,68(-67,+148)
- 5. A-GCG Q8FBQ3:110 agg phi=-60.0+/-0.0 psi=+146.6+/-0.0 (n=1) 2FT0:A,111(-60,+147)